

BUSINESS REQUIREMENTS DOCUMENT

School Transportation Solution

Professionalized Requirements Specification - Version 2

Document Type	Business Requirements Document (BRD)
Solution	School Transportation Management Solution
Target Client Segment	Private schools that operate and fund their own student transportation services
Primary Users	Super Admin, Admin, Transportation Supervisor, Bus Driver, Bus Attendant, Parent/Student
Scope	Student location management, transport feasibility & registration, route planning, multi-trip fleet scheduling, live tracking, attendance, notifications, cost management, and payments

1. Purpose

The purpose of the School Transportation Solution is to provide the school administration with a centralized platform for planning, operating, monitoring, and managing student transportation. The solution is intended to improve route efficiency, student safety, attendance visibility, parent communication, and transportation cost management.

The solution shall support the following business objectives:

- Capture accurate student home locations using a shared map pin.
- Verify transportation feasibility and seat availability, and confirm pricing, before finalizing a student's transportation registration.
- Optimize initial and daily school transportation routes based on home locations, with options focused on shorter travel time and/or lower fuel consumption.
- Support multiple morning and afternoon trips operated by the same bus fleet within a single day.
- Help bus drivers identify and remember each student's home location and assigned stop.
- Provide the Transportation Supervisor with real-time visibility of bus locations to support student safety and protect the school's reputation.
- Track student attendance on each bus through punch-in and punch-out records.
- Allow daily routes to be adjusted based on student attendance and transportation status.
- Provide parents with estimated bus arrival times for student pickup and drop-off.
- Support proactive transportation cost management through a configurable fuel-cost calculator.
- Collect transportation fees online through a payment gateway and support installment-based payments.

1.1 Target Market

The primary target clients for this solution are **private schools** that operate and fund their own student transportation services as a paid offering to parents. All requirements in this document — including feasibility checks, cost management, and online fee collection — are designed around the private-school operating model.

2. Solution Overview

The solution will consist of a backend administration platform and a mobile application. The backend will be used by administrative and transportation management users, while the mobile application will be used by bus drivers, transportation supervisors, and parents/students as applicable.

The solution shall include the following integration capabilities:

- API integration with the school's existing system, where available, to support the exchange of relevant data.
- Integration with a Bluetooth-enabled device installed on each bus to record student punch-in and punch-out activity.
- Integration with WhatsApp as the primary automated channel for parents to share student home-location pins.
- Integration with mapping services, including Google Maps, to support location capture and route navigation.
- Integration with a payment gateway for online collection of transportation fees.
- Generation and distribution of operational reports required to manage school transportation.

3. Functional Requirements

3.1 Student Location and Profile Management

Parents shall be requested to share the student's home location by dropping a pin on Google Maps and sending it to the school through WhatsApp.

- WhatsApp-based pin sharing is the ideal onboarding channel: it enables a fully automated, low-friction registration flow and produces highly accurate location data, since the pin reflects the parent's actual device location rather than a re-typed address. **[NEW]**
- The system shall automate the intake of WhatsApp-shared pins into the student database wherever technically feasible. Manual data entry shall be avoided or eliminated as the default path, and used only as an exception-handling flow, because manually entered data carries a higher risk of inconsistency and human error and is therefore considered lower quality. **[NEW]**
- The Admin user shall be able to upload and label the received location pins to create and maintain the student database.
- The student profile shall support, at minimum, the following information: student name, parent telephone number, gender, class, grade, and home location.
- The system should allow additional fields to be configured or added by the Admin when required.

3.2 Transportation Feasibility, Seat Availability, and Registration Confirmation

New section [NEW]

- Before a student's transportation registration is confirmed, the system shall verify whether the student's home location falls within a serviceable transportation zone/route.
- The system shall check seat availability on the bus(es) that could serve the student's location and request a shift (morning/afternoon) before confirming registration.
- The system shall calculate and present the applicable transportation price based on the feasibility and capacity check results, prior to final registration confirmation.

- Where the location is not serviceable or no seats are available, the system shall notify the relevant user and support alternatives such as waitlisting, reassignment to a different route/bus, or manual review by the Transportation Supervisor.
- Feasibility and seat-availability checks shall be re-evaluated whenever a student's registration, home location, or shift assignment changes.

3.3 Student Transportation Registration and Trip Planning

- The system shall maintain a list of students registered for school transportation services.
- The system shall organize students into trips based on an Admin-selected planning mode.
- Route planning shall support optimization based on shorter travel time/duration and/or lower fuel consumption.
- Route planning shall consider bus passenger capacity, student home location, student grade, student gender, and other criteria configured by the Admin.
- The system shall generate a trip plan showing the sequence of student stops, trip start and end times, and the planned pickup/drop-off time for each student.
- The Transportation Supervisor shall be able to review and modify generated trip plans and customize them based on operational requirements.

3.4 Multiple Daily Trips and Fleet/Bus Scheduling

New section — critical requirement [NEW]

- The system shall support scheduling multiple trips per bus per day, including multiple morning trips and multiple afternoon trips operated by the same bus and driver.
- The system shall detect and prevent scheduling conflicts, ensuring a bus/driver/attendant is not assigned to overlapping trips, and that sufficient turnaround time is allotted between consecutive trips.
- Each trip — regardless of how many trips a given bus runs per day — shall be individually planned, tracked, and reported, with its own stop sequence, timing, attendance, and live-tracking data.
- The Transportation Supervisor shall be able to view and manage a bus's full daily schedule across all of its morning and afternoon trips.
- Route/trip optimization (Section 3.3) shall treat the number of available trips per bus per shift as a capacity constraint when assigning students to trips.

3.5 Driver Trip Management and Navigation

- Each driver shall have access to the trip details assigned to them.
- Trip details shall be printable and/or shareable online in PDF format.
- Where the school provides a tablet to the driver, the system should support step-by-step trip monitoring and navigation using Google Maps.
- The driver shall be able to view the assigned student sequence and relevant trip information during the journey.

- Where a bus operates multiple trips in a day, the driver shall be able to view and switch between the schedule and details for each assigned trip. [NEW]

3.6 Bus Attendant Management

New section [NEW]

- The system shall support a Bus Attendant role: an operational user who rides the bus alongside the driver to assist with student boarding, seating, on-board safety, and punch-in/punch-out activities.
- Bus Attendants shall have access to the mobile application to view their assigned trip(s), the student list/sequence, and relevant trip information.
- Bus Attendants should be able to perform or assist with student punch-in/punch-out, particularly where the driver is unable to do so while driving.
- Bus Attendants shall be able to communicate with the Transportation Supervisor through the same in-app chat/communication functionality available to drivers (Section 3.7).
- The Admin/Transportation Supervisor shall be able to assign one or more Bus Attendants to a specific bus and/or trip.

3.7 Real-Time Bus Tracking and Communication

- The Transportation Supervisor shall be able to monitor the movement of all buses from the backend in real time.
- The system shall display the current bus location to support operational monitoring and student safety.
- The Transportation Supervisor and bus driver shall be able to communicate through an in-app chat function available through the mobile application and backend.
- Chat interactions should be recorded within the system.
- The Transportation Supervisor shall be able to issue common announcements to all bus drivers when required.
- Drivers shall be able to send messages or report operational issues to the Transportation Supervisor through the communication channel.

3.8 Student Attendance and Bus Punch-In/Punch-Out

- The system shall integrate with a device installed on each bus to record student attendance.
- Each student shall be provided with a card/key or equivalent identifier used to punch in when boarding the bus and punch out when leaving the bus.
- Punch-in/punch-out activity may be recorded or assisted by either the Bus Driver or the Bus Attendant, where an attendant is assigned to the trip. [NEW]
- Attendance/punch records shall be tracked separately for the morning session and the afternoon session, supporting cases where a student's transportation status differs between the two. [NEW]

- Each trip shall maintain an attendance log identifying the students who punched in and out.
- The attendance history shall be available to authorized administrative users for monitoring and reporting.

3.9 Absence Management and Dynamic Route Adjustment

The school administration shall be able to record a student's transportation status **independently for the morning session and the afternoon session**, since a student's mode of transportation (e.g., riding the bus vs. an alternative arrangement) or absence may differ between the two sessions. **[UPDATED]**

- The system shall make each session's transportation status (morning/afternoon) available proactively to relevant users.
- The driver or attendant shall be able to adjust the daily route by skipping a stop when a student is absent or is not using the bus for that specific session.
- Route changes resulting from student attendance or transportation status should be reflected in relevant trip and arrival-time information, on a per-session basis.

3.10 Parent Notifications and Estimated Arrival Time

- Parents shall have access to live or updated information regarding the expected bus arrival time for student pickup and drop-off.
- The system shall communicate changes to the expected pickup or drop-off time when the trip schedule changes.
- Arrival-time updates should reflect operational circumstances such as student absence or trip delays.
- Parents shall be able to notify the school/driver separately for the morning session and the afternoon session when the student will not use the bus for that specific session. **[UPDATED]**

3.11 Fuel Cost Calculation and Transportation Cost Management

The system shall provide a configurable fuel-cost calculator based on fuel cost entered by the Admin and the applicable trip details.

- Calculate the fuel cost attributable to a student based on the distance traveled from the student's home to the school through the planned trip stops, multiplied by the applicable fuel cost.
- Calculate the cost using the direct/isolated distance between the student's home and the school, multiplied by the applicable fuel cost.
- Calculate a shared/common trip distance and distribute that distance equally among passengers, then add the applicable individual distance from each student's home to the start of the common route.
- Allow the Admin to configure the applicable calculation approach.

3.12 Online Payment and Installment Management

- The system shall integrate with a payment gateway to collect transportation fees online.
- The system shall issue the required payment/transaction document after successful payment.

- The system shall support installment-based payment of transportation fees.
- Each installment payment shall be recorded against the relevant student.
- The system shall calculate and display the outstanding balance after each payment.

4. User Roles and Access

User Role	Primary Responsibilities
Super Admin	Configures and manages user types and their roles/permissions.
Admin	Manages the transportation solution through the backend, including student data, transportation feasibility/registration, trip planning, operational configuration, cost settings, and payment-related administration.
Transportation Supervisor	Monitors buses in real time, manages and adjusts trips, communicates with drivers, issues announcements, and oversees transportation operations.
Bus Driver	Uses the mobile application to review assigned trip journeys, access student/stop information, communicate with the Transportation Supervisor, and support daily route operations.
Bus Attendant	Uses the mobile application to assist with student boarding, seating, and on-board safety; supports punch-in/punch-out; and communicates with the Transportation Supervisor during assigned trips.
Parent / Student	Uses the mobile application to view bus timing and student bus status, and to notify the school/driver when transportation will not be used.

5. Integration Requirements

Integration	Purpose
School System	API-based integration, where an existing school system is available, for relevant data exchange.
Google Maps	Student home location capture, mapping, and driver navigation.
WhatsApp	Channel through which parents may share dropped home-location pins with the school.
Bluetooth Bus Device	Capture of student punch-in/punch-out events on each bus, per session (morning/afternoon), performed by the driver or bus attendant.

Payment Gateway	Online collection of transportation fees and installment payments.
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6. Reporting Requirements

The system shall provide configurable reports to support transportation administration and operational management. Based on the stated requirements, reporting shall cover, at minimum:

- Student transportation records and profiles.
- Transportation feasibility and seat-availability check outcomes. **[NEW]**
- Trip plans and planned pickup/drop-off schedules.
- Bus/fleet daily trip schedules, including multiple morning and afternoon trips per bus. **[NEW]**
- Bus movement and trip monitoring information.
- Student punch-in/punch-out and attendance records, broken down by morning/afternoon session. **[UPDATED]**
- Student absence and alternative transportation status, per session. **[UPDATED]**
- Fuel and transportation cost calculations.
- Payment transactions, installment collections, and outstanding balances.
- Operational communications and relevant announcements, where required.

7. High-Level Business Process

Updated to reflect feasibility checks, multi-trip scheduling, bus attendants, and per-session status.
[UPDATED]

1. The parent shares the student's home location as a map pin.
2. Admin uploads and labels the location and creates/updates the student profile.
3. The system checks transportation feasibility and seat availability for the student's location and shift, and presents the applicable price, before registration is confirmed.
4. Students registered for transportation are included in the route-planning process, which accounts for the number of trips available per bus per shift (including multiple morning/afternoon trips).
5. The system generates an optimized trip based on the selected planning criteria and configured constraints.
6. The Transportation Supervisor reviews and, where necessary, modifies the trip plan.
7. The driver receives the assigned trip and uses the trip information for daily operations.
8. Students punch in and out using the bus device, assisted by the driver or attendant, creating an attendance record tracked separately for the morning and afternoon sessions.
9. If a student is absent or uses alternative transportation for a given session, that session's status is updated, and the relevant stop may be skipped.
10. The system updates estimated pickup/drop-off timing and makes the information available to parents.

11. The Transportation Supervisor monitors buses in real time and communicates with drivers as required.
12. Fuel and transportation costs are calculated according to the configured methodology.
13. Parents pay transportation fees online, including installments where applicable, and the system records payments and outstanding balances.

8. Assumptions and Dependencies

- Buses may be scheduled for more than one trip per shift (e.g., multiple morning or afternoon runs), and adequate turnaround time between trips is operationally achievable. **[NEW]**
- Bus Attendants, where used, will be provided by the school and equipped with the mobile application. **[NEW]**
- Parents have access to Google Maps and a mechanism such as WhatsApp for sharing the home location pin.
- The school may provide suitable tablets/devices to drivers if step-by-step digital navigation is required.
- A compatible Bluetooth-enabled attendance device will be installed on each bus for punch-in/punch-out integration.
- A payment gateway will be available and configured for transportation fee collection.
- Where school-system integration is required, the school system shall provide suitable APIs and access credentials.
- Accurate route planning depends on the completeness and accuracy of student location and profile data.
- The specific payment provider, Bluetooth device, mapping configuration, fuel-price source/input method, and detailed reporting formats are not specified in the source requirements and therefore require confirmation during detailed solution design.

9. Business-Level Acceptance Criteria

- Authorized users can create and maintain student transportation profiles with accurate home locations.
- The system can determine transportation feasibility and seat availability, and present pricing, before a registration is confirmed. **[NEW]**
- The system can generate a trip plan using the selected route-planning mode and configured constraints.
- A single bus can be scheduled for multiple trips within a shift (e.g., two morning runs) with no scheduling conflicts. **[NEW]**
- The Transportation Supervisor can modify a generated trip before operational use.
- Drivers can access their assigned trip details in printable/PDF form and through the mobile application where applicable.
- A Bus Attendant can access assigned trip information and assist with punch-in/punch-out and communication. **[NEW]**

- The Transportation Supervisor can view bus locations in real time and communicate with drivers.
- Student punch-in/punch-out records are captured and associated with the relevant trip, student, and session (morning/afternoon).
- A student's transportation status and attendance can be recorded and viewed separately for the morning and afternoon session. **[NEW]**
- Student absences or alternative transportation arrangements can be recorded and reflected in daily route operations.
- Parents can view updated expected bus arrival times and relevant student transportation status.
- The fuel-cost calculator supports the business calculation methods described in this BRD.
- Online transportation payments and installments are recorded, and outstanding balances can be determined.

10. Items Requiring Detailed Design Confirmation

- Exact route-optimization algorithm and the definition of the fuel-consumption model.
- Definition of serviceable transportation zones/routes and the exact feasibility-check logic. **[NEW]**
- Bus/driver/attendant scheduling rules and minimum turnaround time between multiple daily trips. **[NEW]**
- Bus Attendant role permissions and the exact division of responsibilities versus the driver. **[NEW]**
- Detailed data model for tracking per-session (morning/afternoon) transportation status and attendance. **[NEW]**
- Real-time tracking hardware/GPS architecture and connectivity requirements.
- Bluetooth attendance device specification, communication protocol, and card/key technology.
- Notification channels and message templates for parents.
- Payment gateway provider, supported payment methods, currency, refund handling, and payment failure scenarios.
- Detailed API scope and data mapping for the school system.
- Detailed report layouts, filters, export formats, and retention requirements.
- User authentication, password policy, and role-based permissions beyond the roles identified in this BRD.